

# Context-Aware Digital Twin for Polymer-Joint Cobots

## Step-by-Step Research & Implementation Roadmap

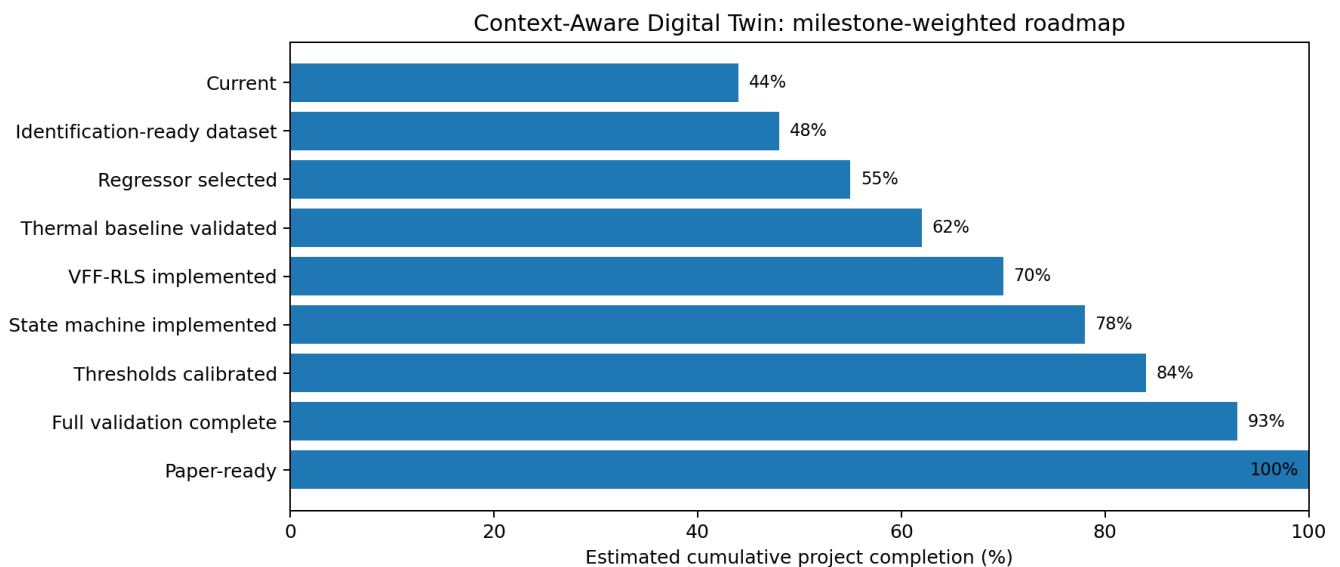
Repository: [dungnguyen15213/6DOF-Rebel-Cobot-Control](#)

Robot: 6-DOF igus ReBeL using CRI/API-level telemetry

Roadmap audit date: 25 September 2026

|                                                   |                                                |                                                   |
|---------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| <b>44%</b><br>Overall project-to-paper completion | <b>~70%</b><br>Engineering/platform foundation | <b>~25%</b><br>Core research novelty + validation |
|---------------------------------------------------|------------------------------------------------|---------------------------------------------------|

**Interpretation:** The robot communication, GUI, logging, baseline FF-RLS, CUSUM, payload prototype, and latency infrastructure provide a strong platform. The main unfinished work is the scientific contribution: final regressor identification, true VFF-RLS, context-state logic, calibration, and experimental validation. Percentages are milestone-weighted planning estimates, not measured labor hours.



### 1. Executive status

**Current decision:** Do not modify the live Digital Twin RLS/state machine yet. First make the repository identification-ready, collect a clean normal-operation dataset, and compare candidate

current regressors offline. Only after the regressor is selected should VFF-RLS and the context-aware state machine be implemented.

**Immediate next milestone:** Identification-ready dataset and offline Model A / Model B comparison.

## 2. What has already been done

- CRI communication and STATUS telemetry pipeline are established.
- PyQt-based Digital Twin / GUI foundation exists and launches from main.py.
- Path planning/export/execution infrastructure exists; the controller owns timing, acceleration, and velocity shaping.
- Diagnostic logging exists and already records time, 6 joint angles, 6 motor currents, 6 velocities, and 6 accelerations.
- Fixed-forgetting-factor RLS exists with  $\lambda = 0.98$  and a 3-parameter configuration.
- Two-sided CUSUM baseline exists with configurable  $k$  and  $h$ .
- A basic payload estimator exists with filter, sensitivity gain, and attach threshold configuration.
- Latency methodology has been separated conceptually into CRI ACK RTT and telemetry-observed motion response.
- Research safety framing is established: supervisory Digital Twin only; no equivalence claim to an external force/torque sensor.
- VFF-RLS mathematics V1 and context-state logic V1 have now been defined conceptually, but are not yet implemented or calibrated.

## 3. What is not yet complete

- Final physically defensible current regressor  $\phi(q, \dot{q}, \ddot{q})$ .
- Experimental decision on whether reconstructed acceleration improves held-out current prediction.
- Cross-axis / multi-DOF gravity coupling model justified from robot geometry or data.
- True residual-dependent VFF-RLS in code.
- Robust residual normalization and slow residual-energy statistic for VFF control.
- Adaptation freeze/gating during collision, soft-contact, or pending payload decisions.
- Context-aware state machine implementation and hysteresis / timing rules.
- Stationary threshold, grasp context, soft-contact threshold, collision threshold, and CUSUM threshold calibration.
- Payload sensitivity and empty-state calibration at planned masses 0 / 100 / 250 / 500 g.
- Cold-vs-warm collision/contact experiments and final confusion matrices / ROC / precision / recall / F1.
- Final payload RMSE / MAE / bias / standard deviation.
- Final paper results and conclusions.

## 4. Repository audit for the next step

**Verdict:** YES - the public repository is suitable as a starting point for the next model-identification step, with a few preparation checks. There is no need to rebuild the GUI, CRI layer, or basic logger before beginning offline identification.

| Repo evidence                    | What it shows                                                                             | Readiness | Action                                                                                      |
|----------------------------------|-------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|
| config.yaml                      | FF-RLS $\lambda=0.98$ ; $n\_params=3$ ; fixed CUSUM $k/h$ ; simple payload gain/threshold | Partial   | Keep as baseline; do not call it VFF.                                                       |
| settings.py                      | Typed settings support only current FF-RLS, CUSUM, payload, logging parameters            | Partial   | Later add VFF/state-machine settings after equations/calibration are frozen.                |
| robot_diagnostics.csv            | Time, $q1..q6$ , $current1..6$ , $velocity1..6$ , $acceleration1..6$ are present          | Good      | Use this schema as the base identification dataset.                                         |
| robot_diagnostics.csv timestamps | Sampling intervals are non-uniform in the checked log                                     | Important | Always use measured $\Delta t$ ; never assume fixed sampling for differentiation/filtering. |
| QUICK_START / PATH_EXPORT_GUIDE  | Path execution exists and controller performs motion timing/profile shaping               | Good      | Use repeatable commanded trajectories for identification; log actual STATUS telemetry.      |
| main.py                          | Application entry point and GUI wiring are straightforward                                | Good      | No GUI work needed for the immediate model-identification milestone.                        |

**Repository limitation of this audit:** The public GitHub root and raw top-level files were accessible, but GitHub directory-tree crawling was restricted in this session. Therefore this audit verifies the top-level configuration, startup, path-planning documentation, and checked diagnostic output, but it does not claim a line-by-line review of every file under core/, gui/, hardware/, and tests/.

## 5. Small repository preparation before data collection

**5.1 Preserve raw telemetry** - Confirm the experimental CSV contains the raw CRI STATUS joint position/current fields before any RLS or anomaly-processing transformation. Keep processed signals in additional columns rather than overwriting raw values.

**5.2 Use real timestamps** - Use monotonic/sample timestamps and compute  $dt[k] = t[k] - t[k-1]$ . The checked CSV is not uniformly sampled.

**5.3 Add run metadata** - At minimum record experiment\_id, trajectory\_id, run\_id, commanded speed, payload\_g, contact\_label, grasp\_context, and cold/warm condition. For the first identification dataset set payload\_g=0 and contact\_label=NORMAL.

**5.4 Freeze live analytics during identification** - Do not use adaptive RLS output as the target. Fit the offline model against measured motor current. Existing FF-RLS/CUSUM can continue running for display, but they must not alter the logged raw telemetry.

**5.5 Keep units explicit** - Current log uses degrees, deg/s, deg/s<sup>2</sup>, and mA. Convert to a consistent modeling convention before fitting, preferably rad, rad/s, rad/s<sup>2</sup>, and A, while retaining original columns.

**5.6 Treat existing acceleration as unverified** - The CSV contains acceleration, but before using it in Model B, compare it against a filtered derivative of velocity computed with actual dt. Do not assume the current acceleration column is sufficiently low-noise for identification.

## 6. Step-by-step TODO list

### STEP 1 - Make an identification-ready logger/dataset

*Status: NEXT | Timing: 1-2 lab sessions*

- Create a dedicated experiment output file instead of reusing an arbitrary diagnostics file.
- Required columns: timestamp, q1..q6, qdot1..qdot6, current1..current6. Keep qddot1..qddot6 as optional/candidate columns.
- Add run\_id / trajectory\_id / condition metadata.
- Verify no missing values, duplicated timestamps, unit ambiguity, or current clipping.
- Confirm actual dt distribution from timestamps.
- Definition of done: one CSV can be loaded offline and each sample can be assigned to a specific trajectory/run.

### STEP 2 - Collect clean normal-operation identification data

*Status: NEXT | Timing: 1-2 lab sessions*

- No payload, no intentional contact, no collision.
- Record positive and negative motion for all joints.

- Use low, medium, and higher safe commanded speeds.
- Include single-joint excitation and representative multi-joint trajectories.
- Repeat trajectories as separate runs so entire runs can be held out for validation.
- Definition of done: at least several independent runs with sufficient motion diversity and clean labels.

### STEP 3 - Build offline preprocessing

*Status: AFTER STEP 2 | Timing: Notebook/script*

- Convert units to A and radians-based kinematics.
- Use measured dt.
- Filter qdot before numerical differentiation.
- Generate a candidate qddot signal with a documented filter/differentiator.
- Plot q, qdot, qddot, current, and dt for quality control.
- Definition of done: preprocessing is deterministic and reproducible from raw CSV.

### STEP 4 - Fit Model A

*Status: AFTER STEP 3 | Timing: Offline*

- Use  $\phi_A = [1, \dot{q}_i, \tanh(\dot{q}_i/v_s), \sin(q_i), \cos(q_i)]$ .
- Fit parameters per joint with numerically stable least squares (not explicit matrix inversion).
- Compute train and held-out RMSE, MAE, residual mean/std, and regression condition number.
- Definition of done: Model A has per-joint validation metrics and residual plots.

### STEP 5 - Fit Model B and decide whether acceleration stays

*Status: AFTER STEP 4 | Timing: Offline*

- Use  $\phi_B = [1, \ddot{q}_i, \dot{q}_i, \tanh(\dot{q}_i/v_s), \sin(q_i), \cos(q_i)]$ .
- Use the same train/validation split as Model A.
- Keep acceleration only if it produces meaningful held-out improvement without unstable/noisy parameters.
- Definition of done: evidence-based A-vs-B decision for each joint and one chosen baseline regressor.

### STEP 6 - Investigate multi-DOF gravity/coupling terms

*Status: AFTER STEP 5 | Timing: Model design*

- Use actual ReBeL kinematics/geometry to propose cross-axis gravity terms; do not add arbitrary sine combinations.
- Add only terms that are physically interpretable and identifiable from the data.
- Repeat held-out comparison against the selected simpler model.
- Definition of done: final regressor  $\phi$  is frozen for all later estimator comparisons.

## STEP 7 - Thermal/warm-up baseline experiment

*Status: AFTER REGRESSOR FREEZE | Timing: ~45 min experiment*

- Repeat a controlled trajectory from cold start through warm operation.
- Keep the selected static model fixed first.
- Measure residual drift versus elapsed time/temperature proxy.
- Definition of done: quantitative evidence that the baseline changes enough to motivate adaptation.

## STEP 8 - Implement context-gated VFF-RLS

*Status: AFTER STEP 7 | Timing: Code + replay test*

- Implement normalized a-priori residual, robust slow residual-energy statistic,  $\lambda_{\min}/\lambda_{\max}$  law, and  $\lambda$  smoothing.
- Implement adaptation gate so collision/contact candidates freeze parameter updates.
- Retain fixed-FF RLS as a baseline selectable by config.
- Definition of done: static, fixed-FF, and VFF estimators can replay the same dataset and produce comparable outputs.

## STEP 9 - Implement the context-aware state machine

*Status: AFTER VFF | Timing: Code + replay test*

- States: NORMAL/ADAPTIVE DRIFT, TRANSIENT COLLISION, SOFT CONTACT, PAYLOAD/GRASP EVENT.
- Payload classification permitted only under stationary + grasp context.
- Add persistence timers, release thresholds, hysteresis, and event priorities.
- Definition of done: deterministic state transitions can be tested from prerecorded logs.

## STEP 10 - Calibration experiments

*Status: AFTER STATE MACHINE | Timing: Lab*

- Estimate residual scale  $\sigma_e$  per joint from normal data.
- Calibrate stationary velocity thresholds, CUSUM parameters, collision/soft-contact thresholds, timing windows, and VFF bounds/time constants.
- Calibrate payload sensitivity using 0/100/250/500 g and empty-state references.
- Freeze thresholds using development data only.
- Definition of done: all thresholds in config are experimentally justified and frozen before final test.

## STEP 11 - Final validation experiments

*Status: AFTER CALIBRATION | Timing: Lab*

- Static baseline vs fixed-FF RLS vs VFF-RLS.
- Cold and warm collision trials.
- Sustained soft-contact trials.
- Payload/grasp trials with planned masses.
- Compute ROC/precision/recall/F1/confusion matrices and payload RMSE/MAE/bias/std.
- Definition of done: test-set results are complete and no threshold is retuned on final test data.

## STEP 12 - Manuscript integration

*Status: FINAL* | *Timing: Writing*

- Update Section 3 with final regressor + VFF equations.
- Update Section 4 with exact state-machine guards/transitions.
- Report latency datasets separately unless explicitly correlated.
- Use the phrase sensorless contact and payload inference without an external force/torque sensor.
- State clearly that the Digital Twin augments, not replaces, certified safety functions.
- Definition of done: every technical claim maps to implemented equations and an experiment/result.

## 7. Progress matrix

| Workstream                             | Weight | Done credit | Workstream completion | Status          |
|----------------------------------------|--------|-------------|-----------------------|-----------------|
| Research scope & safety framing        | 5%     | 5%          | 100%                  | Complete/strong |
| CRI / STATUS / GUI foundation          | 15%    | 14%         | 93%                   | Complete/strong |
| Logging & latency infrastructure       | 10%    | 8%          | 80%                   | Complete/strong |
| Baseline FF-RLS / CUSUM / payload      | 10%    | 7%          | 70%                   | In progress     |
| Final regressor / model identification | 15%    | 3%          | 20%                   | In progress     |
| Context-gated VFF-RLS                  | 12%    | 3%          | 25%                   | In progress     |
| Context state machine                  | 12%    | 3%          | 25%                   | In progress     |
| Calibration experiments                | 8%     | 0%          | 0%                    | Not started     |
| Validation experiments                 | 8%     | 0%          | 0%                    | Not started     |

|                                   |    |    |     |             |
|-----------------------------------|----|----|-----|-------------|
| Results & manuscript finalization | 5% | 1% | 20% | In progress |
|-----------------------------------|----|----|-----|-------------|

**Overall milestone-weighted completion = 44%.** This number intentionally gives substantial weight to the still-unfinished scientific novelty and validation. It should be updated after each milestone rather than treated as a precise productivity metric.

## 8. Exact next actions - do these now

1. Do not change the live RLS formula yet.
2. Create/confirm a dedicated identification CSV logger with raw time,  $q$ ,  $\dot{q}$ , current and run metadata.
3. Run one short dry test and inspect timestamp spacing, units, missing data, and current saturation.
4. Collect the first no-payload/no-contact trajectory set with separate run IDs.
5. Create an offline notebook/script that loads those runs and constructs Model A.
6. Add filtered acceleration and Model B only after Model A works.
7. Return with the dataset/notebook outputs; then freeze the regressor before VFF implementation.

## 9. Planned repository changes after the regressor is frozen

These are intentionally deferred so Copilot/code changes do not drive the research design:

- Extend config.yaml/settings.py with VFF bounds/time constants, residual normalization settings, gating thresholds, and state-machine timers.
- Add a model/regressor module with a stable interface for static, fixed-FF, and VFF estimators.
- Add context/state-machine module independent from the GUI.
- Add replay-based unit/integration tests using recorded telemetry.
- Add experiment metadata and versioned calibration files.
- Keep raw telemetry logs immutable; write derived analytics to separate outputs.

## 10. Repository evidence checked

- Repository root: <https://github.com/dungnguyen15213/6DOF-Rebel-Cobot-Control>
- config.yaml: <https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/config.yaml>
- settings.py: <https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/settings.py>
- main.py: <https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/main.py>
- QUICK\_START.md: [https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/QUICK\\_START.md](https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/QUICK_START.md)

- PATH\_EXPORT\_GUIDE.md: [https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/PATH\\_EXPORT\\_GUIDE.md](https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/PATH_EXPORT_GUIDE.md)
- PATH\_EXPORT\_REFERENCE.md: [https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/PATH\\_EXPORT\\_REFERENCE.md](https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/PATH_EXPORT_REFERENCE.md)
- robot\_diagnostics.csv: [https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/robot\\_diagnostics.csv](https://raw.githubusercontent.com/dungnguyen15213/6DOF-Rebel-Cobot-Control/main/robot_diagnostics.csv)

## 11. Research guardrails to keep throughout the project

- Do not claim that the Digital Twin replaces the E-stop or certified safety controller.
- Do not claim equivalence to a force/torque sensor.
- Use: sensorless contact and payload inference without an external force/torque sensor.
- Do not call RTT/2 measured one-way latency; call it estimated one-way communication delay under an approximate symmetry assumption.
- Do not call telemetry-observed motion response pure actuator latency.
- Do not let Copilot invent the VFF law, regressor physics, state-machine guards, or calibration thresholds.
- Do not tune final thresholds on the test set.